

Final Project Proposal: Global Supply Chain Resilience Architect

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Project Overview:

- **Selected Title:** Global Supply Chain Resilience Architect
- **Target Audience:** Suppliers, manufacturers, warehouses, and distributors
- **Core Objective:** To create a dashboard to visualize five years of shipping and logistics data and illustrate bottlenecks and the relationship between geopolitical events and lead times.

Data Acquisition Plan:

- **Primary Source:**
 - o UNCTAD Merchandise Trade Total (UNCTADstat — US.TradeMerchTotal)
Annual global merchandise trade statistics compiled by the United Nations Conference on Trade and Development, covering total import and export values across countries and regions. Data is expressed in current US dollars and updated regularly via the UNCTADstat Data Centre.
 - o Geopolitical Risk (GPR) Index — Caldara & Iacoviello (matteoiacoviello.com/gpr.htm) A news-based measure of adverse geopolitical events and associated risks, constructed by tallying newspaper articles covering geopolitical tensions, with data examining its evolution and economic effects since 1900. The Benchmark GPR Index draws from 10 major newspapers and started in 1985, with country-specific GPR indexes also available for advanced and emerging economies.
- **Scope:** Five years of annual/monthly statistics (2020-2024) from UNCTAD on global merchandise trade flows (imports/exports by country/region), combined with monthly GPR index values for the same time.
- **Estimated Data Size:**
 - o UNCTAD dataset: Approximately 5,000 to 15,000 rows depending on country-year-flow combinations selected; CSV/Excel export from UNCTADstat
 - o GPR dataset: 60 monthly observations per index (2020–2024); available as a clean Excel file directly from the Iacoviello website
- **Wrangling Challenge:**
 - o Temporal alignment: UNCTAD trade data is annual, whereas GPR data is weekly; aggregation or interpolation will be required to align them on a single time axis.

- o Missing values: UNCTAD data may contain gaps for smaller economies or years with delayed reporting, which needs imputation or exclusion judgments.

Analytical Framework:

- **Key Question 1:** Do geopolitical events affect global shipping?
- **Key Question 2:** What is the pearson coefficient between geopolitical events and lead times?

Toolset

- **Language:** Python
- **Environment:** Jupyter Notebook
- **Libraries:**
 - o Python pandas
 - o Matplotlib
 - o Dash